

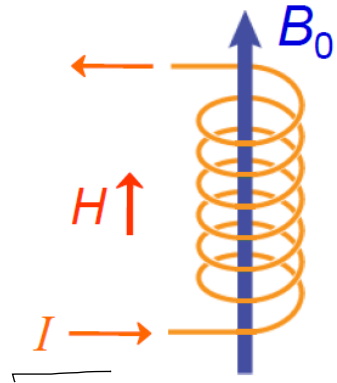
Chapter 20: Magnetic Properties

ISSUES TO ADDRESS...

- What are the important magnetic properties?
- How do we explain magnetic phenomena?
- How are magnetic materials classified?
- How does magnetic memory storage work?
- What is superconductivity and how do magnetic fields effect the behavior of superconductors?

Generation of a Magnetic Field -- Vacuum

- Created by current through a coil:



N = total number of turns

ℓ = length of each turn (m)

I = current (ampere)

H = applied magnetic field (ampere-turns/m)

B_0 = magnetic flux density in a vacuum (tesla)

- Computation of the applied magnetic field, H :

$$H = \frac{NI}{\ell}$$

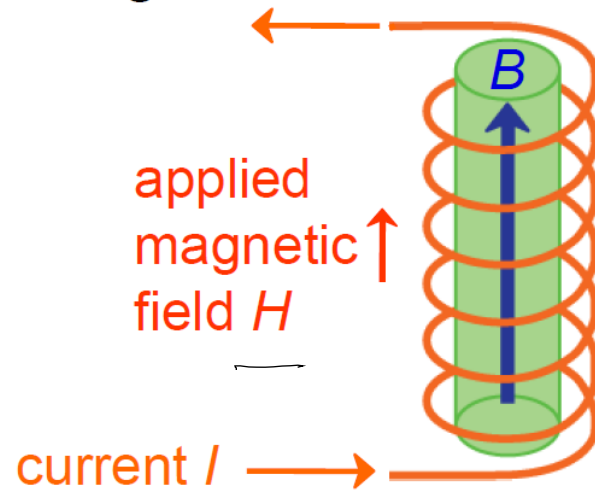
- Computation of the magnetic flux density in a vacuum, B_0 :

$$B_0 = \mu_0 H$$

permeability of a vacuum
(1.257×10^{-6} Henry/m)

Generation of a Magnetic Field -- within a Solid Material

- A magnetic field is induced in the material



B = Magnetic Induction (tesla)
inside the material

$$B = \mu H$$

permeability of a solid

$$B_0 = \mu_0 H$$

$$H = \frac{NI}{l}$$

- Relative permeability (dimensionless)

$$\mu_r = \frac{\mu}{\mu_0}$$

$$\mu = \mu_r \cdot \mu_0$$

Generation of a Magnetic Field -- within a Solid Material (cont.)

- Magnetization

$$M = \chi_m H$$

Magnetic susceptibility
(dimensionless)

- B in terms of H and M

$$B = \mu_0 H + \mu_0 M$$

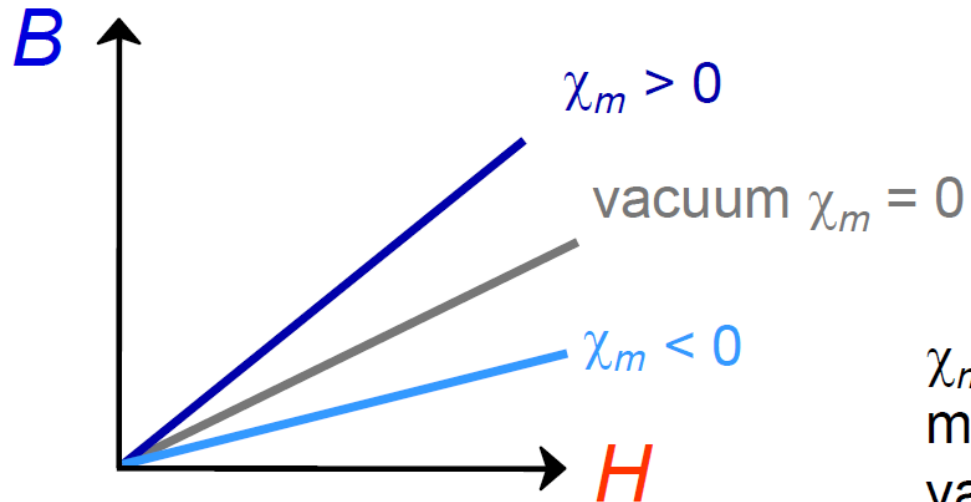
- Combining the above two equations:

$$B = \mu_0 H + \mu_0 \chi_m H$$

$$= (1 + \chi_m) \mu_0 H$$

permeability of a vacuum:
(1.26×10^{-6} Henry/m)

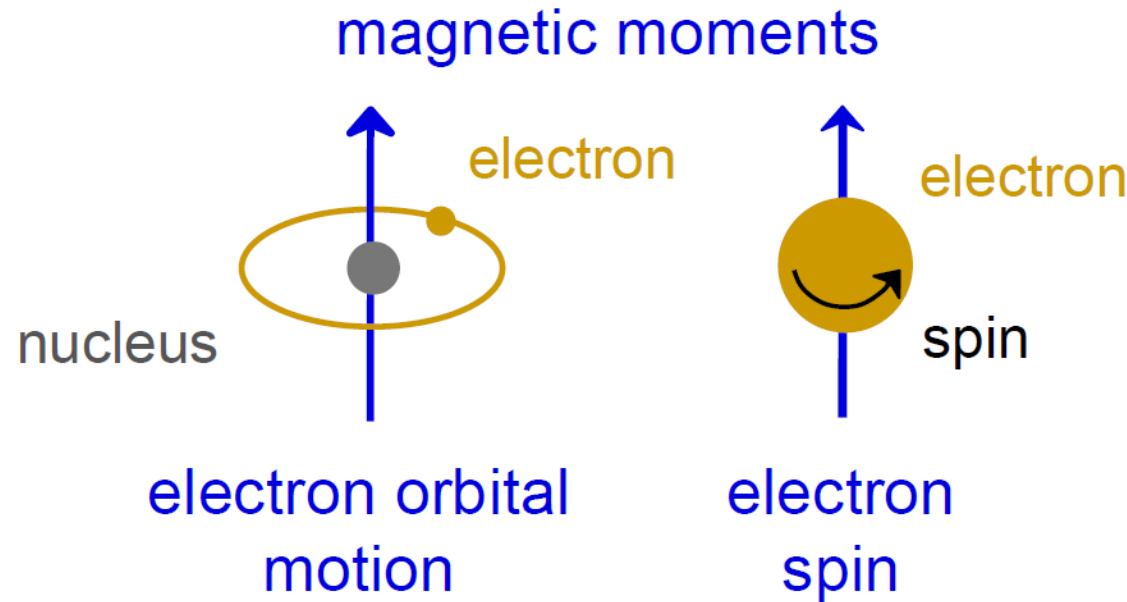
$$\mu_r = 1 + \chi_m$$



χ_m is a measure of a material's magnetic response relative to a vacuum

Origins of Magnetic Moments

- Magnetic moments arise from electron motions and the spins on electrons.



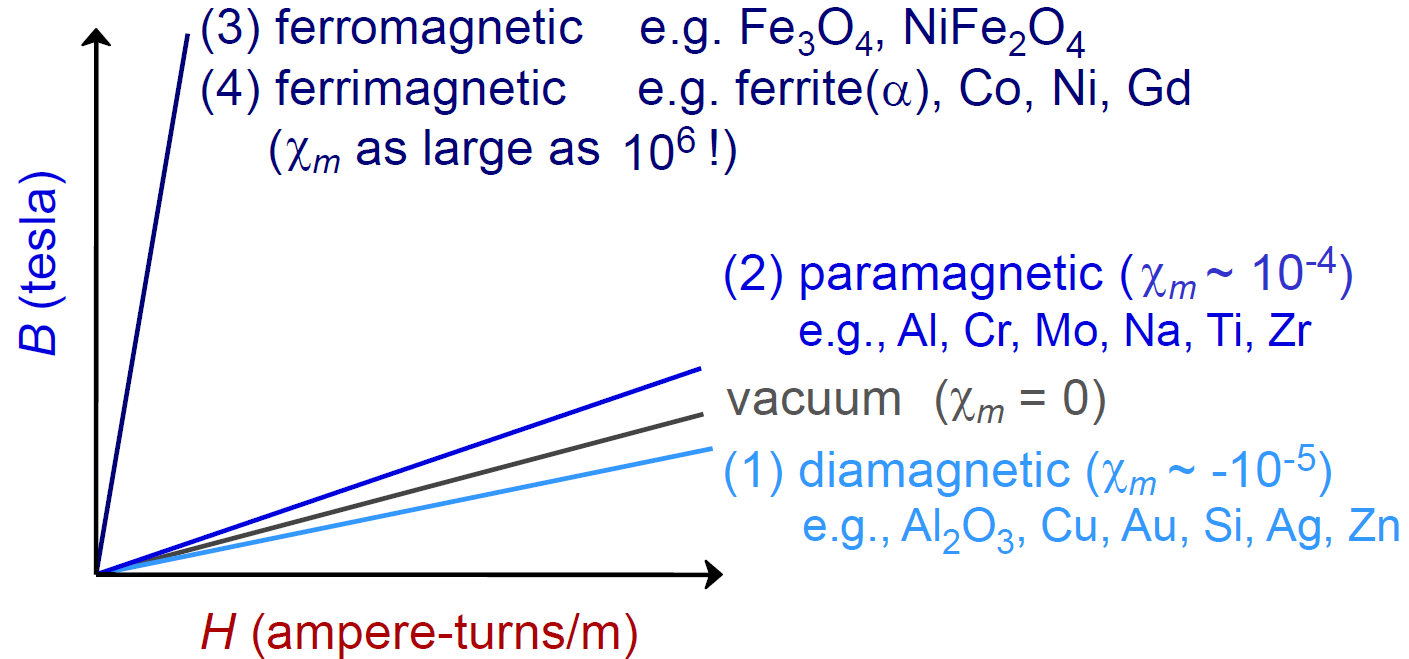
Adapted from Fig. 20.4,
Callister & Rethwisch 8e.

- Net atomic magnetic moment:
-- sum of moments from all electrons.
- Four types of response...

Types of Magnetism

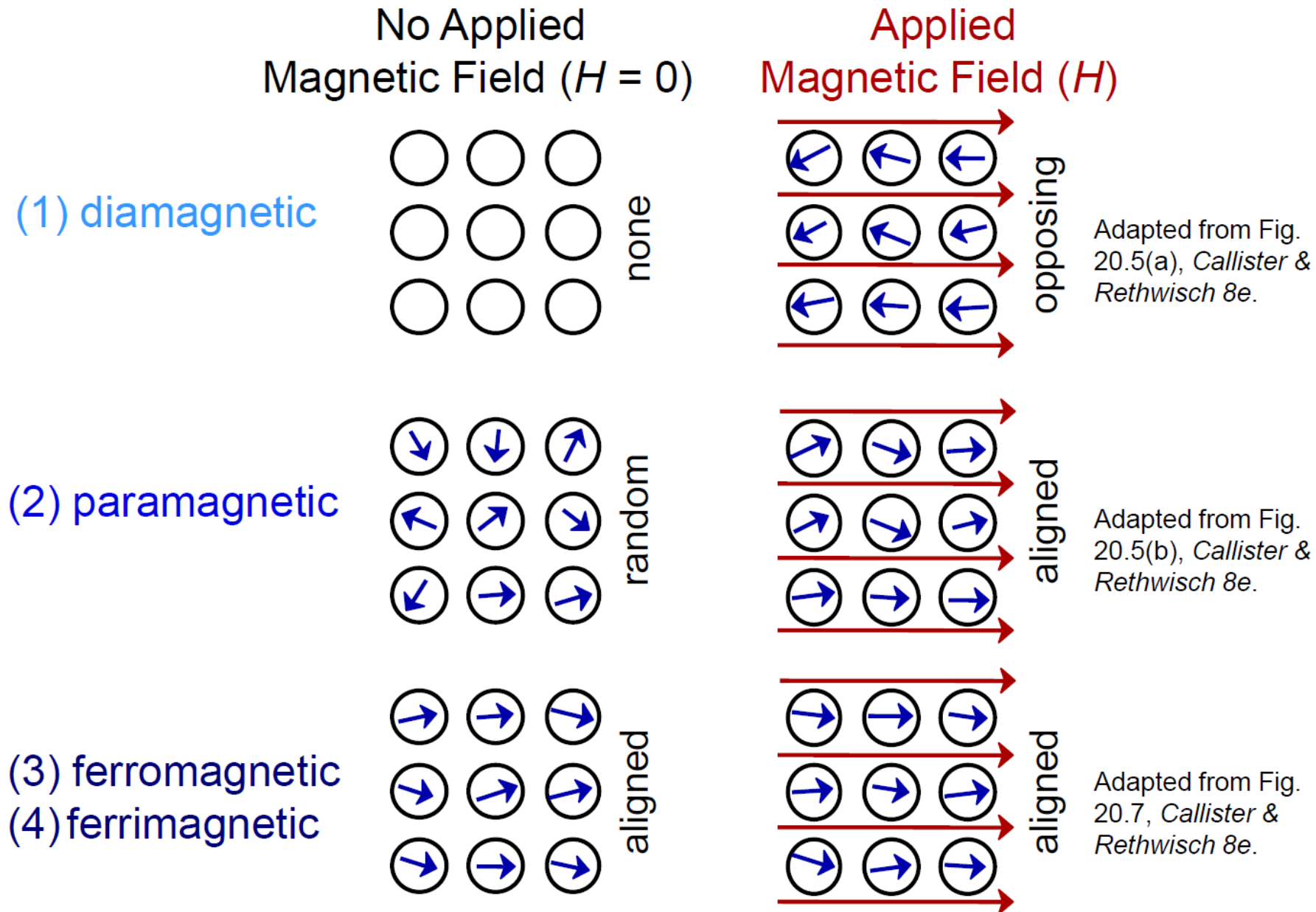
$$B = (1 + \chi_m) \mu_0 H$$

$$B = \mu_0 H$$



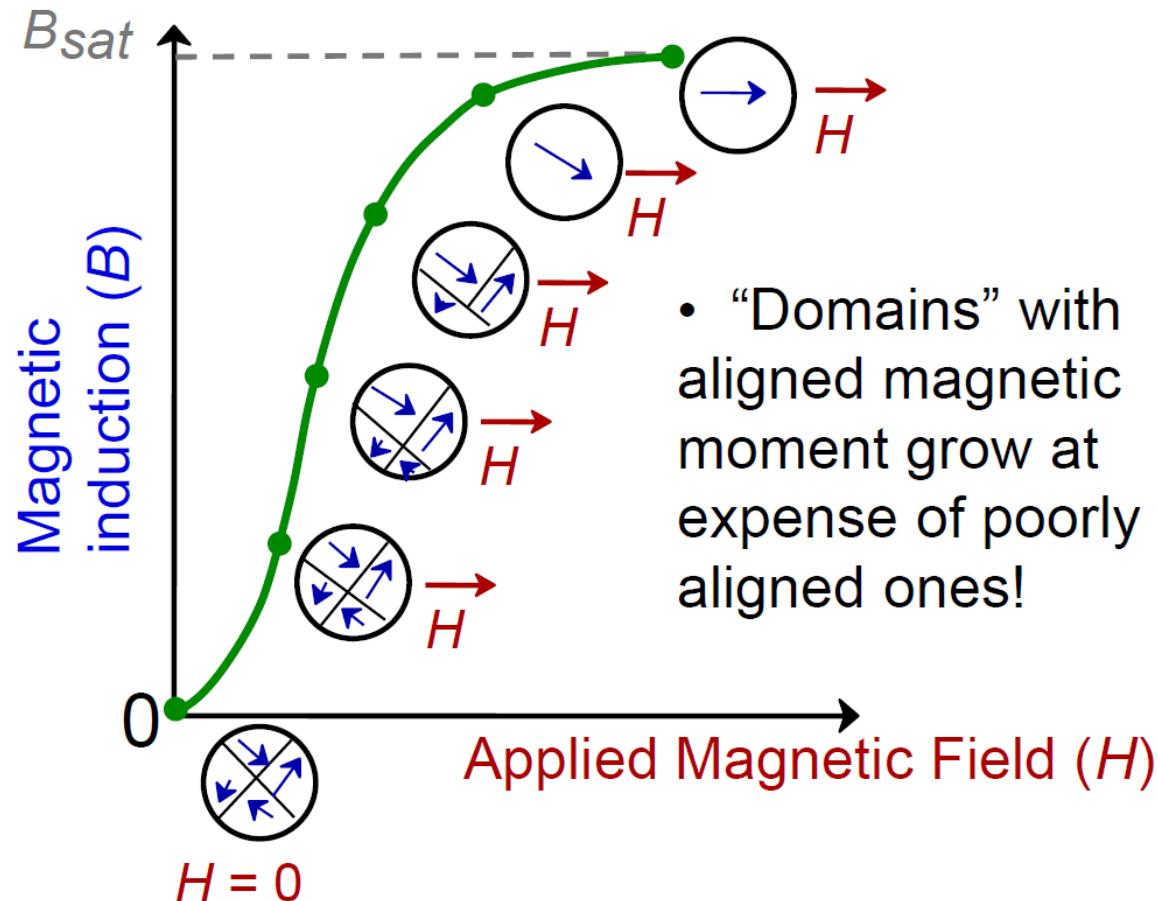
<i>Diamagnetics</i>		<i>Paramagnetics</i>	
<i>Material</i>	<i>Susceptibility</i> χ_m (volume) (SI units)	<i>Material</i>	<i>Susceptibility</i> χ_m (volume) (SI units)
Aluminum oxide	-1.81×10^{-5}	Aluminum	2.07×10^{-5}
Copper	-0.96×10^{-5}	Chromium	3.13×10^{-4}
Gold	-3.44×10^{-5}	Chromium chloride	1.51×10^{-3}
Mercury	-2.85×10^{-5}	Manganese sulfate	3.70×10^{-3}
Silicon	-0.41×10^{-5}	Molybdenum	1.19×10^{-4}
Silver	-2.38×10^{-5}	Sodium	8.48×10^{-6}
Sodium chloride	-1.41×10^{-5}	Titanium	1.81×10^{-4}
Zinc	-1.56×10^{-5}	Zirconium	1.09×10^{-4}

Magnetic Responses for 4 Types



Domains in Ferromagnetic & Ferrimagnetic Materials

- As the applied field (H) increases the magnetic domains change shape and size by movement of domain boundaries.



Adapted from Fig. 20.13, Callister & Rethwisch 8e. (Fig. 20.13 adapted from O.H. Wyatt and D. Dew-Hughes, *Metals, Ceramics, and Polymers*, Cambridge University Press, 1974.)

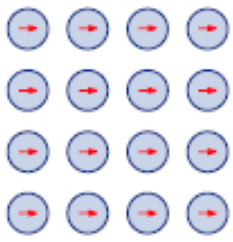


Table 20.4 Net Magnetic Moments for Six Cations

Cation	Net Spin Magnetic Moment (Bohr magnetons)
Fe ³⁺	5
Fe ²⁺	4
Mn ²⁺	5
Co ²⁺	3
Ni ²⁺	2
Cu ²⁺	1

Saturation Magnetization and Flux Density Computations for Nickel

Calculate (a) the saturation magnetization and (b) the saturation flux density for nickel, which has a density of 8.90 g/cm^3 .

Solution

(a) The saturation magnetization is just the product of the number of Bohr magnetons per atom (0.60 as given earlier), the magnitude of the Bohr magneton μ_B , and the number N of atoms per cubic meter, or

$$M_s = 0.60\mu_B N \quad (20.9)$$

Now, the number of atoms per cubic meter is related to the density ρ , the atomic weight A_{Ni} , and Avogadro's number N_A , as follows:

$$\begin{aligned} N &= \frac{\rho N_A}{A_{\text{Ni}}} \quad (20.10) \\ &= \frac{(8.90 \times 10^6 \text{ g/m}^3)(6.022 \times 10^{23} \text{ atoms/mol})}{58.71 \text{ g/mol}} \\ &= 9.13 \times 10^{28} \text{ atoms/m}^3 \end{aligned}$$

Finally,

$$\begin{aligned} M_s &= \left(\frac{0.60 \text{ Bohr magneton}}{\text{atom}} \right) \left(\frac{9.27 \times 10^{-24} \text{ A}\cdot\text{m}^2}{\text{Bohr magneton}} \right) \left(\frac{9.13 \times 10^{28} \text{ atoms}}{\text{m}^3} \right) \\ &= 5.1 \times 10^5 \text{ A/m} \end{aligned}$$

(b) From Equation 20.8, the saturation flux density is just

$$\begin{aligned} \text{---} \circ B_s &= \mu_0 M_s \\ &= \left(\frac{4\pi \times 10^{-7} \text{ H}}{\text{m}} \right) \left(\frac{5.1 \times 10^5 \text{ A}}{\text{m}} \right) \\ &= 0.64 \text{ tesla} \end{aligned}$$

Bohr magneton

Ni $\rightarrow 0.6$

$$M_s = 0.6 \times \mu_B \times N$$

Saturation Magnetization Determination for Fe_3O_4

Calculate the saturation magnetization for Fe_3O_4 given that each cubic unit cell contains 8 Fe^{2+} and 16 Fe^{3+} ions, and that the unit cell edge length is 0.839 nm.

Solution

This problem is solved in a manner similar to Example Problem 20.1, except that the computational basis is per unit cell as opposed to per atom or ion.

The saturation magnetization will be equal to the product of the number N' of Bohr magnetons per cubic meter of Fe_3O_4 , and the magnetic moment per Bohr magneton μ_B ,

$$M_s = N' \mu_B \quad (20.11)$$

Now, N' is just the number of Bohr magnetons per unit cell n_B divided by the unit cell volume V_C , or

$$N' = \frac{n_B}{V_C} \quad (20.12)$$

Again, the net magnetization results from the Fe^{2+} ions only. Because there are 8 Fe^{2+} ions per unit cell and 4 Bohr magnetons per Fe^{2+} ion, n_B is 32. Furthermore, the unit cell is a cube, and $V_C = a^3$, a being the unit cell edge length. Therefore,

$$\begin{aligned} \propto M_s &= \frac{n_B \mu_B}{a^3} \\ &= \frac{(32 \text{ Bohr magnetons/unit cell})(9.27 \times 10^{-24} \text{ A}\cdot\text{m}^2/\text{Bohr magneton})}{(0.839 \times 10^{-9} \text{ m})^3/\text{unit cell}} \\ &= 5.0 \times 10^5 \text{ A/m} \end{aligned} \quad (20.13)$$

$$B_s = \mu_0 M_s$$

Table 20.3 The Distribution of Spin Magnetic Moments for Fe^{2+} and Fe^{3+} Ions in a Unit Cell Fe_3O_4 ^a

Cation	Octahedral Lattice Site	Tetrahedral Lattice Site	Net Magnetic Moment
Fe^{3+}	↑ ↑ ↑ ↑ ↑ ↑ ↑ ↑	↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓	Complete cancellation
Fe^{2+}	↑ ↑ ↑ ↑ ↑ ↑ ↑ ↑	—	↑ ↑ ↑ ↑ ↑ ↑ ↑ ↑

^a Each arrow represents the magnetic moment orientation for one of the cations.

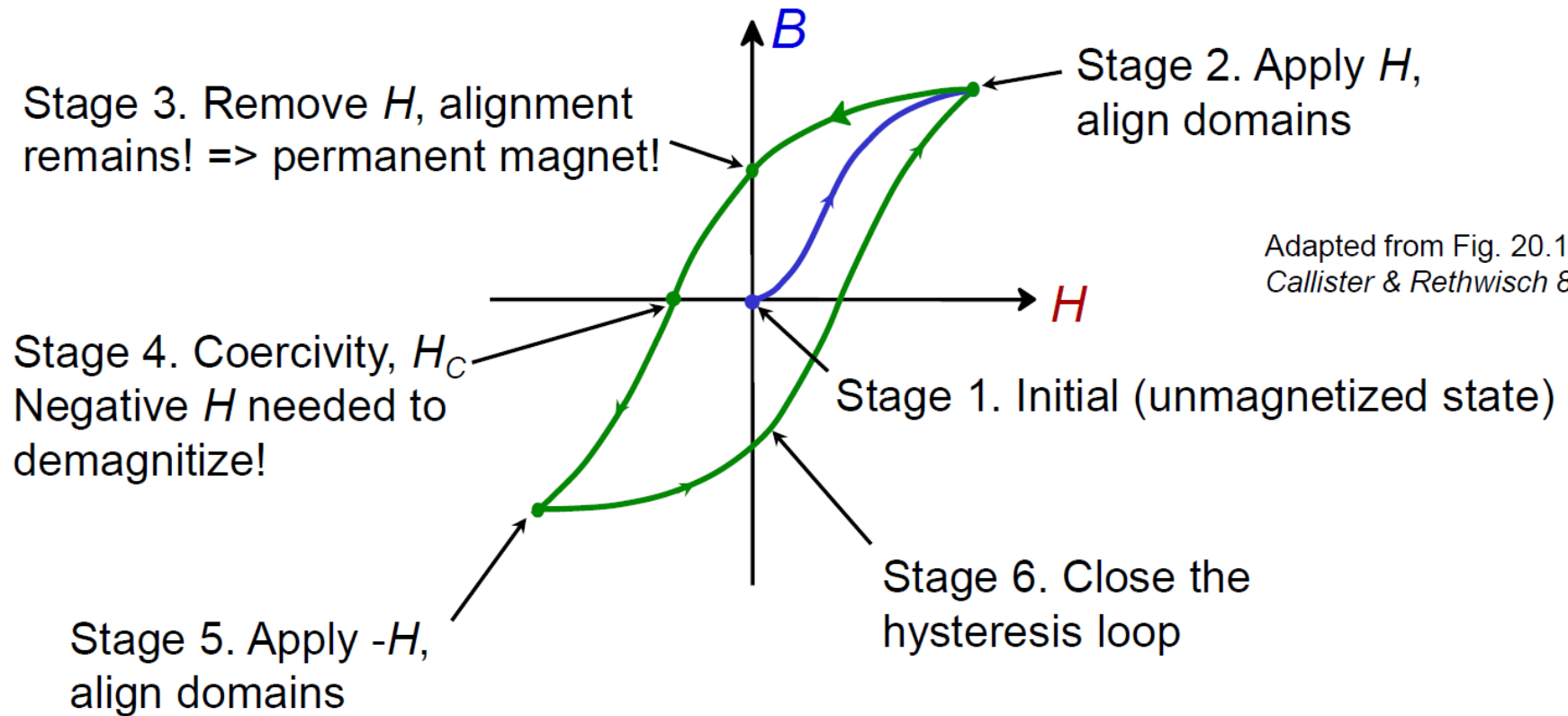
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$$(8 \times 4) \mu_B / a^3$$

Hysteresis and Permanent Magnetization

- The magnetic hysteresis phenomenon



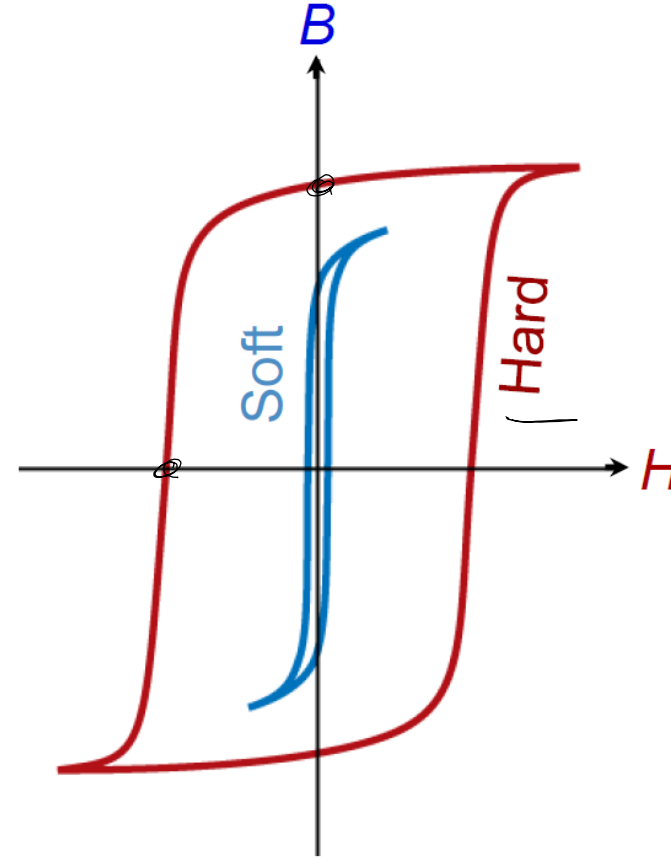
Hard and Soft Magnetic Materials

Hard magnetic materials:

- large coercivities
- used for permanent magnets
- add particles/voids to inhibit domain wall motion
- example: tungsten steel --
 $H_c = 5900$ amp-turn/m)

Soft magnetic materials:

- small coercivities
- used for electric motors
- example: commercial iron 99.95 Fe



Adapted from Fig. 20.19, *Callister & Rethwisch 8e*. (Fig. 20.19 from K.M. Ralls, T.H. Courtney, and J. Wulff, *Introduction to Materials Science and Engineering*, John Wiley and Sons, Inc., 1976.)

Magnetic Storage

- Digitized data in the form of electrical signals are transferred to and recorded digitally on a magnetic medium (tape or disk)
- This transference is accomplished by a recording system that consists of a **read/write head**
 - “**write**” or record data by applying a magnetic field that aligns domains in small regions of the **recording medium**
 - “**read**” or retrieve data from medium by sensing changes in magnetization

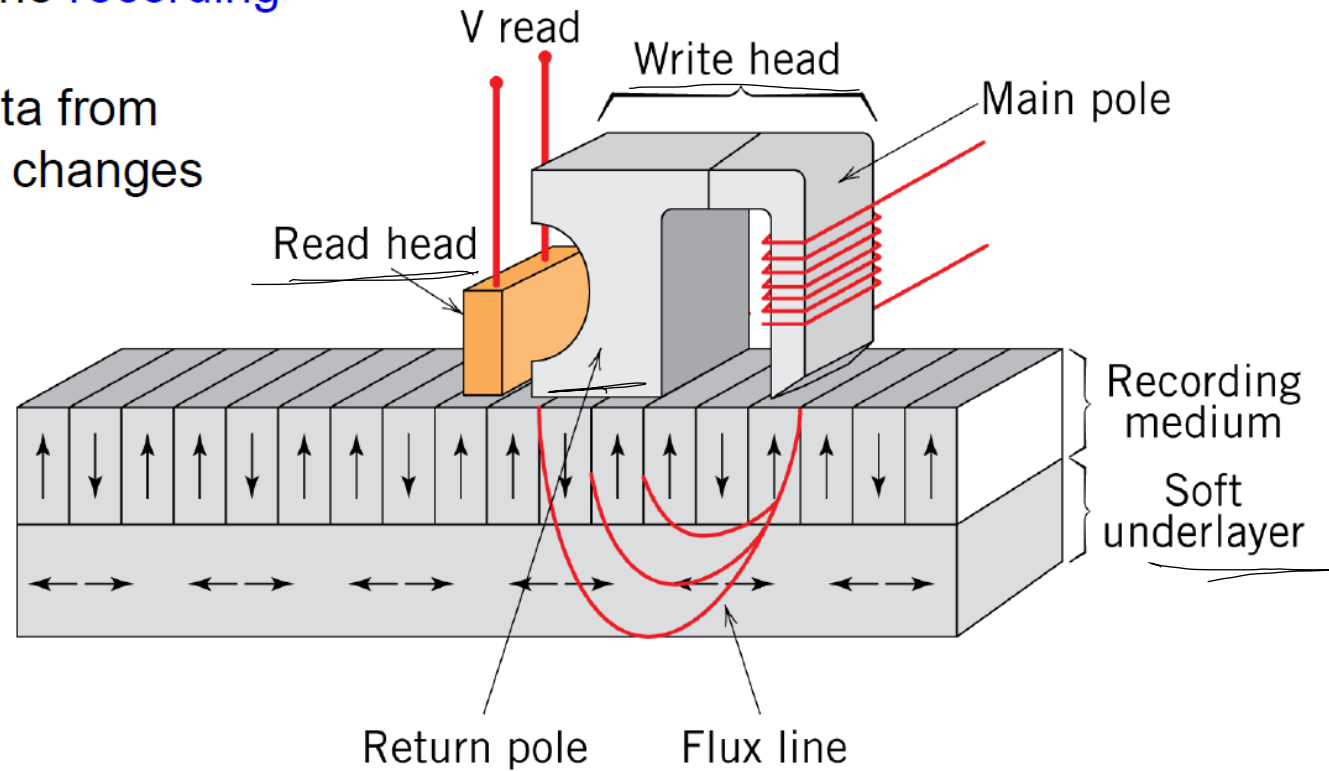


Fig. 20.23, Callister & Rethwisch 8e.

Magnetic Storage Media Types

- Hard disk drives (**granular/perpendicular media**):

- CoCr alloy grains (darker regions) separated by oxide grain boundary segregant layer (lighter regions)
- Magnetization direction of each grain is perpendicular to plane of disk

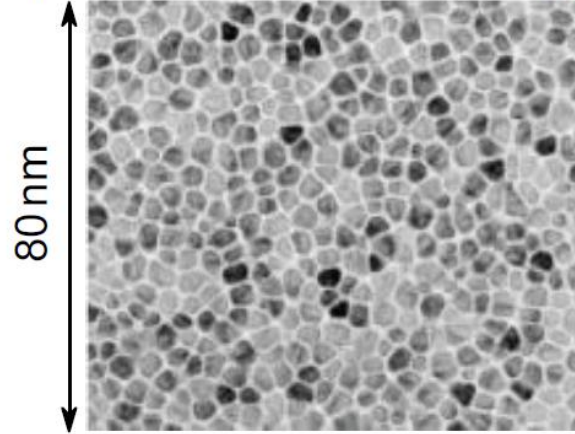
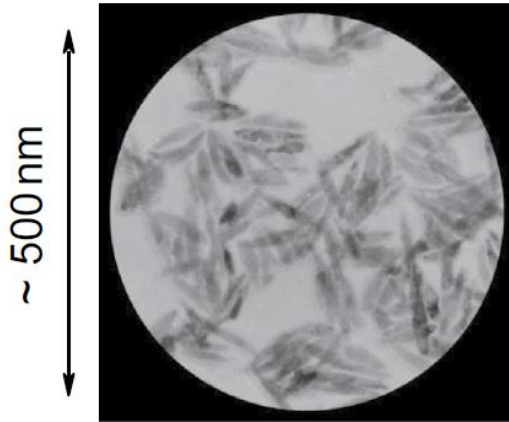
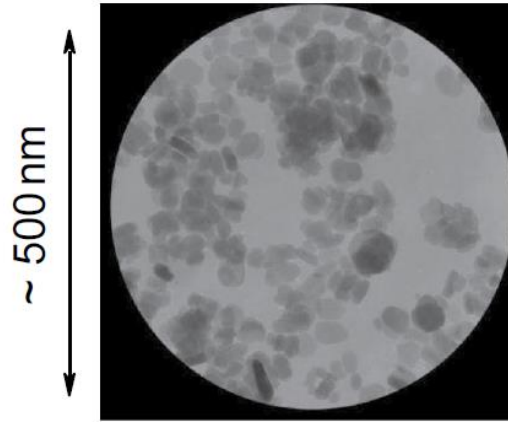


Fig. 20.25, *Callister & Rethwisch 8e.*
(Fig. 20.25 from Seagate Recording Media)

- Recording tape (**particulate media**):



- Acicular (needle-shaped) ferromagnetic metal alloy particles



- Tabular (plate-shaped) ferrimagnetic barium-ferrite particles

Fig. 20.24, *Callister & Rethwisch 8e.*
(Fig. 20.24 courtesy Fuji Film Inc., Recording Media Division)

Superconductivity

Found in 26 metals and hundreds of alloys & compounds

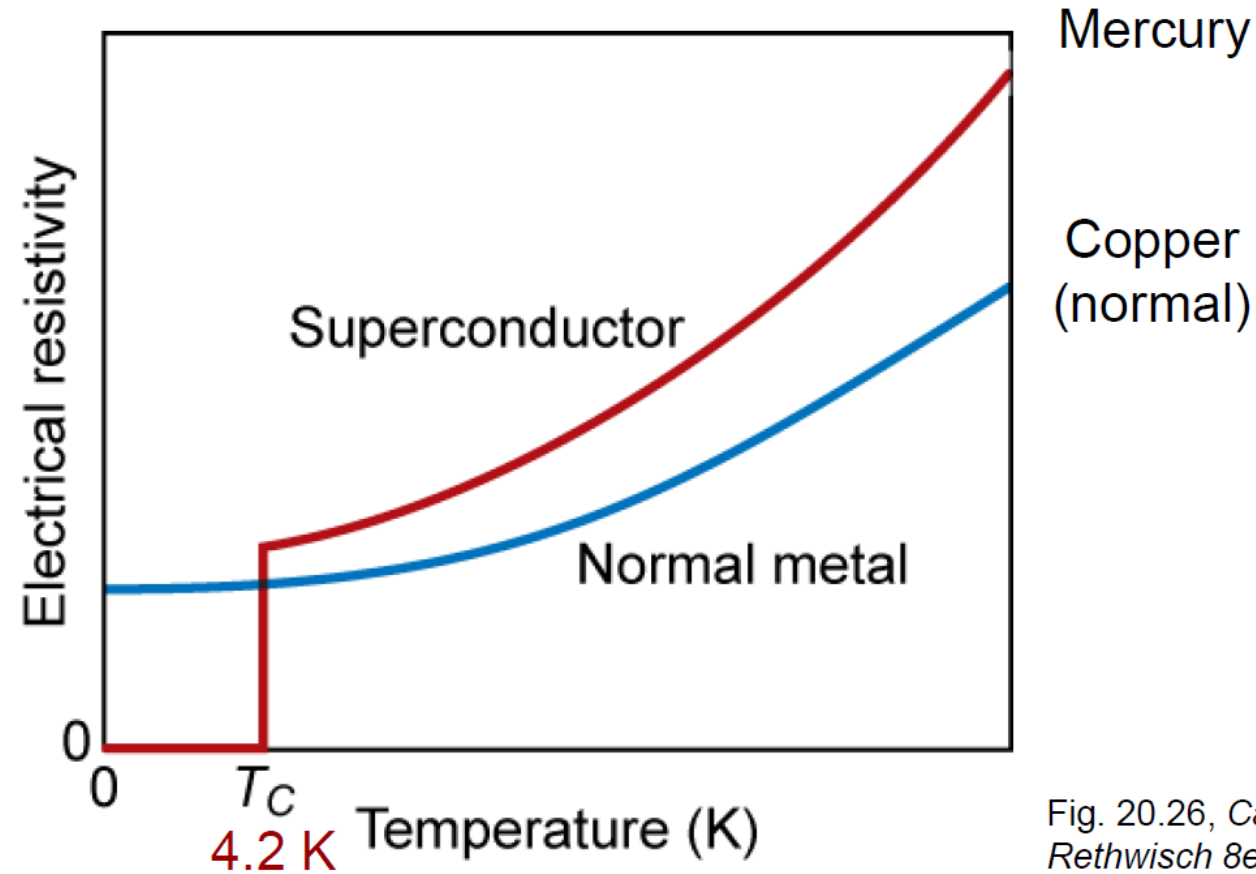
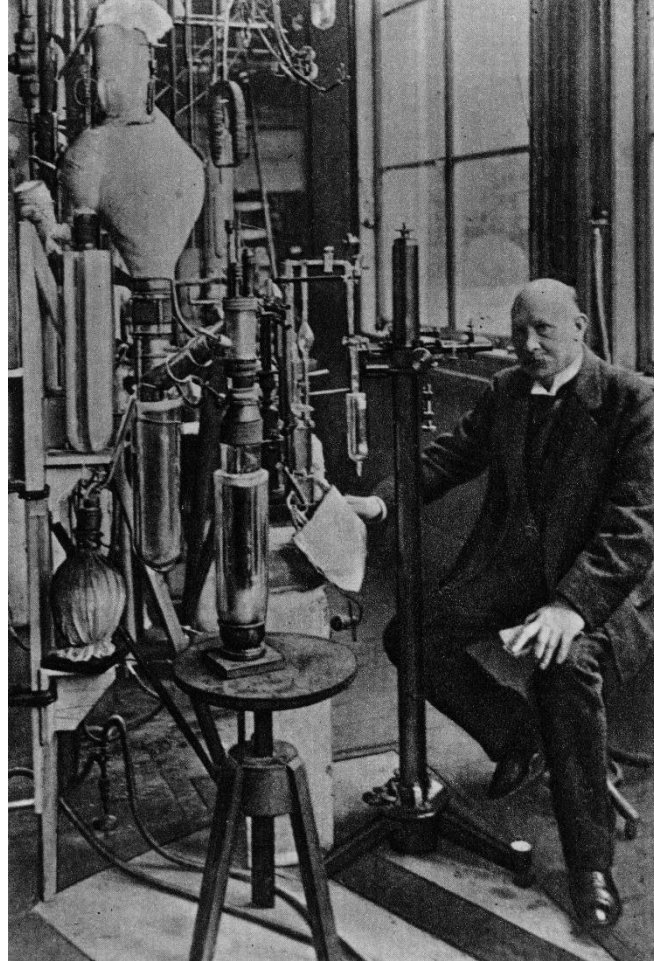


Fig. 20.26, Callister & Rethwisch 8e.

- T_C = critical temperature
= temperature below which material is superconductive

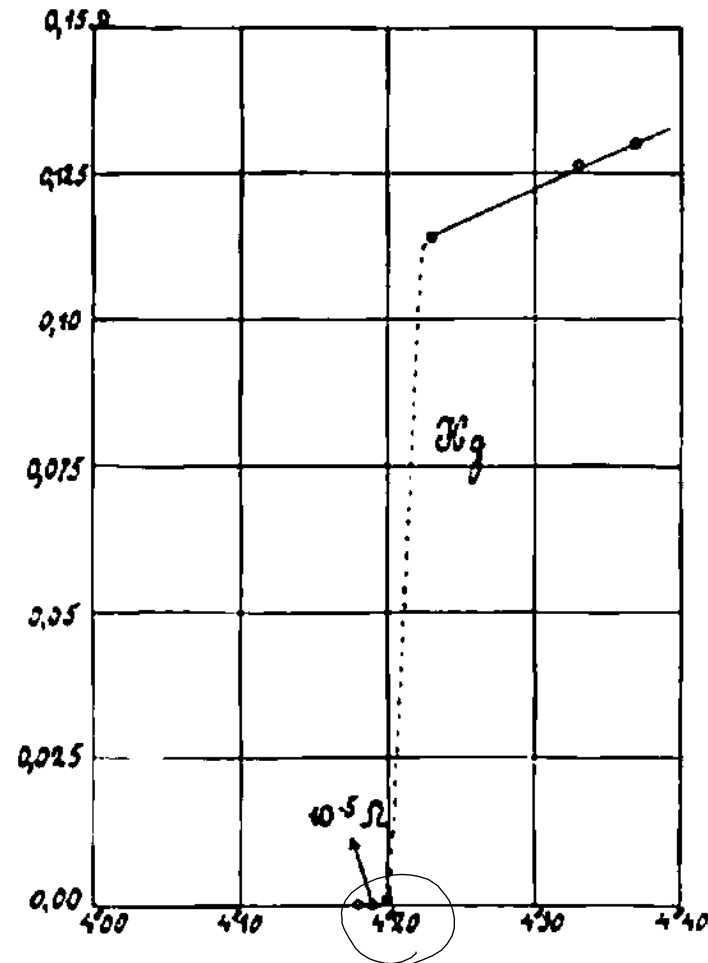
History of Superconductivity

- Heike Kamerlingh Onnes produces liquid He (1908)



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- Heike Kamerlingh Onnes produces liquid He (1908)
- He discovered superconductivity in a Hg wire (1911)

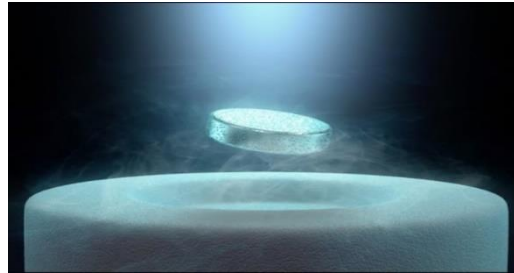


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Superconductivity

- Second order phase transition
- Perfect conductor
- Perfect diamagnet (Meissner effect)

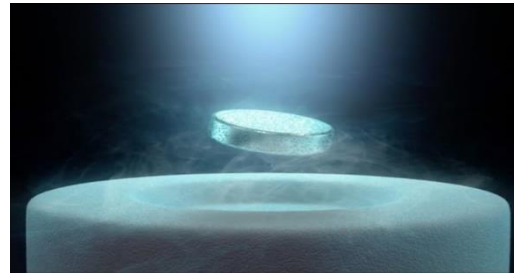
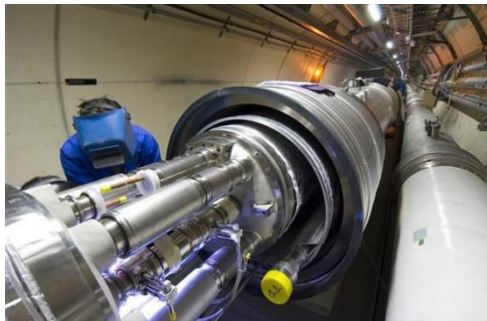


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- Second order phase transition
- Perfect conductor
- Perfect diamagnet (Meissner effect)



Applications

- Fundamental and applied research (\$1B)
Synchrotron and X-ray instruments
- Medical and analytical instrumentation (\$5B)
MRI and NMR
- Energy technology (\$25B)
Frictionless bearings, energy storage



History of Superconductivity

- Heike Kamerlingh Onnes produces liquid He (1908)
- He discovered superconductivity in a Hg wire (1911)
- Quantum nature of superconductivity stays a mystery for decades
- Bardeen, Cooper and Schrieffer (BCS) proposed their groundbreaking theory for superconductivity (1957)

PHYSICAL REVIEW

VOLUME 108, NUMBER 5

DECEMBER 1, 1957

Theory of Superconductivity*

J. BARDEEN, L. N. COOPER,[†] AND J. R. SCHRIEFFER[‡]
Department of Physics, University of Illinois, Urbana, Illinois
(Received July 8, 1957)



Bardeen, Cooper and Schrieffer (BCS)

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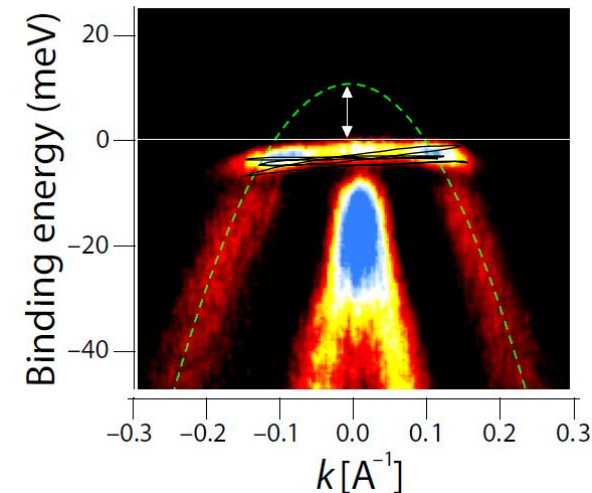
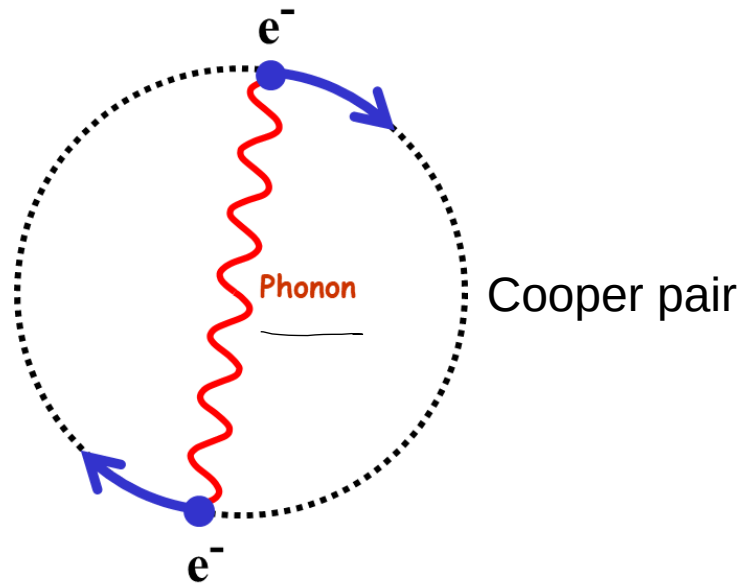
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Rinott et al., Science (2017)

Electrons (fermion) form Cooper pairs (boson)

Critical Properties of Superconductive Materials

T_C = critical temperature - if $T > T_C$ not superconducting

J_C = critical current density - if $J > J_C$ not superconducting

H_C = critical magnetic field - if $H > H_C$ not superconducting

$$H_C(T) = H_C(0) \left(1 - \frac{T^2}{T_C^2} \right)$$

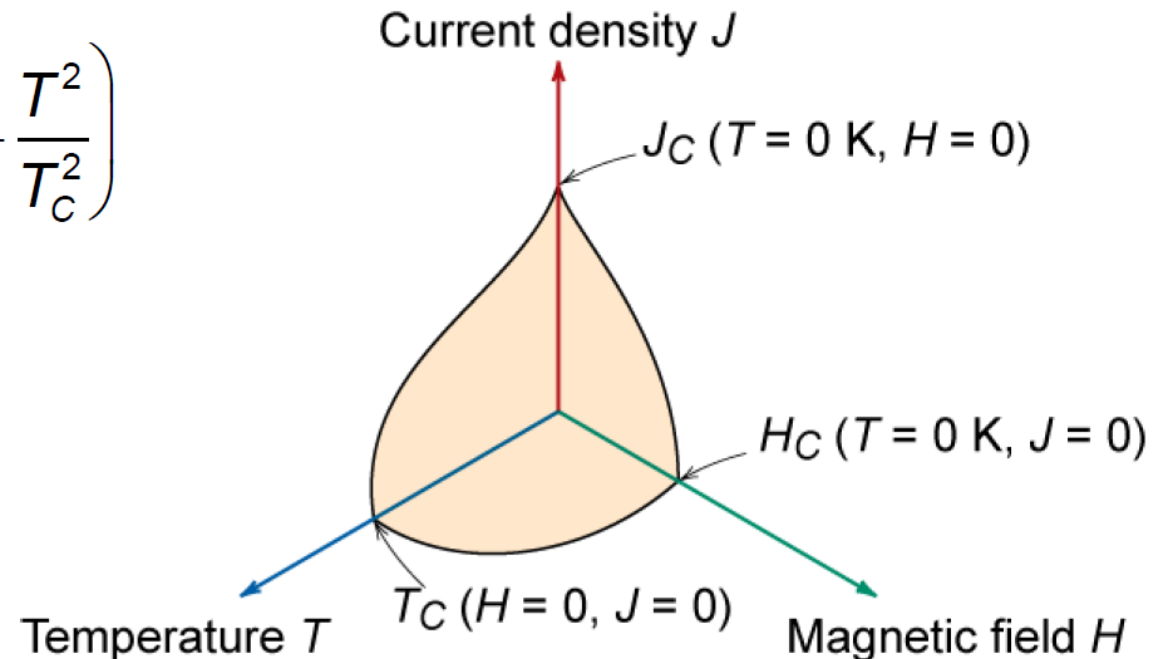
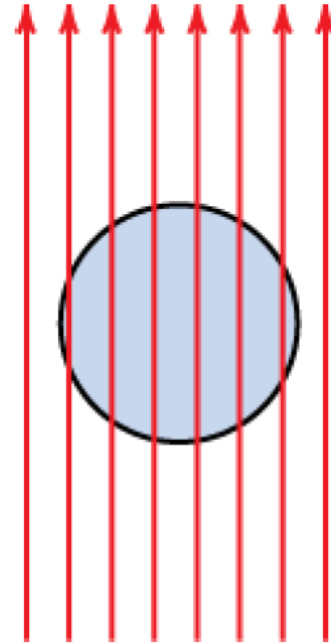


Fig. 20.27, Callister & Rethwisch 8e.

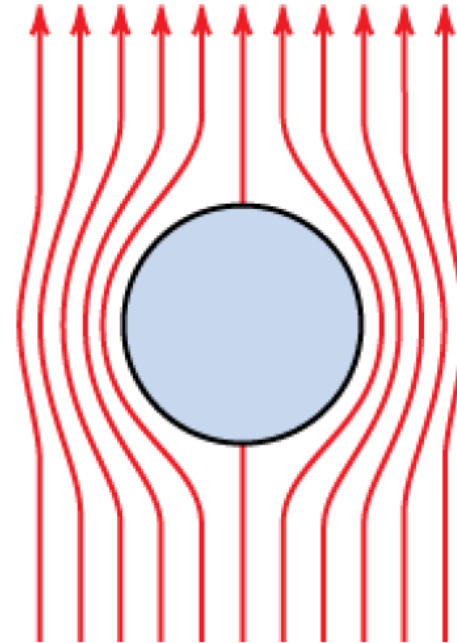
Meissner Effect

- Superconductors expel magnetic fields

$$\chi = -1$$



normal



superconductor

Fig. 20.28, Callister & Rethwisch 8e.

- This is why a superconductor will float above a magnet

Advances in Superconductivity

- Research in superconductive materials was stagnant for many years.
 - Everyone assumed $T_{C,max}$ was about 23 K
 - Many theories said it was impossible to increase T_C beyond this value
- 1987- new materials were discovered with $T_C > 30$ K
 - ceramics of form $Ba_{1-x} K_x BiO_{3-y}$
 - Started enormous race
 - $Y Ba_2 Cu_3 O_{7-x}$ $T_C = 90$ K
 - $Tl_2 Ba_2 Ca_2 Cu_3 O_x$ $T_C = 122$ K
 - difficult to make since oxidation state is very important
- The major problem is that these ceramic materials are inherently brittle.

Summary

- A magnetic field is produced when a current flows through a wire coil.
- **Magnetic induction (B):**
 - an internal magnetic field is induced in a material that is situated within an external magnetic field (H).
 - magnetic moments result from electron interactions with the applied magnetic field
- Types of material responses to magnetic fields are:
 - **ferrimagnetic** and **ferromagnetic** (large magnetic susceptibilities)
 - **paramagnetic** (small and positive magnetic susceptibilities)
 - **diamagnetic** (small and negative magnetic susceptibilities)
- Types of **ferrimagnetic** and **ferromagnetic** materials:
 - **Hard**: large coercivities
 - **Soft**: small coercivities
- Magnetic storage media:
 - particulate barium-ferrite in polymeric film (tape)
 - thin film Co-Cr alloy (hard drive)

$$H = \frac{NI}{l}$$

$$B = \mu H$$

$$B = (1 + \chi) \mu_0 H$$